Paddy Iyer

DATA ENGINEERING LEADER • STRATEGIC DATA ARCHITECT • AI & PRIVACY ENGINEERING • AI INNOVATOR

15+ years driving enterprise data transformation for global tech leaders across cloud/SaaS, fintech, gaming, and consumer platforms. Expert in cloud-native architectures, real-time analytics, privacy-first pipelines, and AI-augmented data operations. Proven builder of high-trust, high-performance data teams at petabyte scale. Published writer and thought leader on data architecture, AI agents, and the intersection of ancient wisdom with modern technology.

Modern Data Architecture · Privacy-First Design · Consent Architecture · AI Agents & LLM Pipelines · Prompt Engineering · Data Governance (C1–C5) · Petabyte-Scale Systems · Global Team Leadership · Data Storytelling

FEATURED EXPERIENCE

Data Engineer — Privacy & Consent Infrastructure *Apr 2024 – Present*

Meta

- Delivered ~$500K/year in recovered engineering capacity by sustaining the output of 3–4 engineers as a single engineer through AI-assisted development — 1,000+ production changes landed
- Eliminated ~52,000 redundant data operations per year, removing an estimated 2–4 FTE of recurring manual toil (~$500K–$1M annually in engineering time, storage, and metadata overhead)
- Achieved this leverage at under $1,000 in total LLM spend — roughly 500x return on inference cost — by optimizing on cost per landed change (~$0.80) rather than cost per API call
- Cut LLM input tokens ~60% via cacheable prompt prefixes and routed simple transforms to zero-cost local engines, holding per-change cost flat as volume scaled
- Cut manual cross-namespace data imports 91% (156.8/day → 14.2/day), driving the final phase of a multi-year warehouse consolidation from 80% to 100% coverage
- Unblocked the last ~20% of warehouse data — the segment no prior effort could touch because it sat in the direct ads revenue path, where a single day of degradation carries seven-figure impact — and delivered it with zero revenue loss
- Reduced engineer ramp-up from 2 weeks to 2 days and per-asset cycle time 3–5x by building a reusable AI agent playbook, with every hard-block gate derived from a production postmortem
- Operated under a constraint where errors surface as revenue loss rather than test failures, which drove the guard-first design: proved the pre-deployment validation stack was structurally blind — permission checks passed for tables that did not exist — and built the system that closed the gap
- Collapsed an entire failure class into an offline-checkable invariant: 81% of migrations required a namespace change, not the name change the tooling assumed
- Owned incident response on revenue-critical infrastructure — 5 production incidents including 2 high-severity — resolved fix-forward with zero reverts and permission grants coordinated across 6+ teams

EARLIER CAREER

Lead Data Architect — Partner Data Engineering *Jan 2022 – Jan 2024*

VMware

Strategic architect for VMware's Partner Data Platform — unifying fragmented partner data into a single governed analytics layer

- Designed and built the Partner Data Platform from the ground up, unifying partner data from 10+ disparate sources into a single governed analytics layer — eliminating silos that had blocked cross-functional decision-making for years
- Implemented VMware's first structured data governance for partner data (classification, lineage, policy enforcement) plus a self-service data catalog — cutting tribal-knowledge reliance by 60% and ad-hoc request volume by ~40%, and achieving audit-ready status

CONTACT

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San Francisco Bay Area

TECHNICAL SKILLS

Spark · Hive · Hadoop · Databricks · Delta Lake · Snowflake · Redshift · dbt · Airflow · Dataswarm · AWS · Azure · GCP · Terraform · Kubernetes · Python · SQL · Scala · LLM/RAG Pipelines · Claude Code · AI Agents · Prompt Eng. · LangChain · GitHub Copilot · MLflow · Diff. Privacy · Unity Catalog · Data Mesh

EDUCATION

Data Engineering Cert.
UC Santa Cruz, 2013–14

B.S. Electrical Engineering
Sardar Patel College of Eng.

Diploma, Electronics & Comms
Technical Board, Tamil Nadu

PUBLICATIONS

- Cloud Computing
- All About Big Data
- Future Trends in BI

AI & THOUGHT LEADERSHIP

- 160+ articles on paddyspeaks.com
- Claude Code skill development
- Vibe coding methodology
- AI agent architecture writing

OPEN-SOURCE & COMMUNITY

- Interview Studio — free Data & AI Engineering prep
- 22 data-model deep-dives + 1,500+ questions
- In-browser SQL / Python (WASM), no backend

EARLIER CAREER

Data Engineer Consultant
LinkedIn
Dec 2017 – Apr 2018
GDPR compliance — PII encryption into Dali storage (Hive, Python, Pig)

Lead Data Engineer
Meta
Sep 2015 – Sep 2017
80+ Dataswarm pipelines for petabyte-scale ads; Cross Device Insights, Outcomes Datamart + Norms DB

Data Engineer
GREE International
2014 – 2015
PII masking, Vertica→Redshift, 1,000+ table optimization

Data Architect
Chegg Inc.
Oct 2013 – Apr 2014

Technical Director
Model N
2011 – 2013
Life Sciences BI, cloud migration, ETL

modularization

Architect
CallidusCloud
2005 – 2011
Incentive comp analytics, BusinessObjects XI

DW Architect
Hewlett Packard
2001 – 2005
Enterprise DW + 8 datamarts; ETL 18→3 hrs; 40–50% sales lift via clickstream

- Pioneered an AI copilot for retrieval optimization, coding, and governance — one of VMware's earliest LLM-assisted data engineering deployments, reducing routine query development time by ~30%
- Achieved 40%+ Spark processing improvements via broadcast joins, caching, and skew management — bringing SLA-critical reports inside their delivery windows for the first time
- Engaged director/VP-level stakeholders to define a unified data vision, securing budget and headcount that grew the team from 3 to 8 engineers during the engagement

Senior Consultant — Ads, Commerce & Privacy *Jul 2019 – Dec 2021*

Meta

Led high-impact projects across Meta's Ads, Commerce, and Privacy teams, architecting secure, scalable, and compliant data platforms

- Led privacy remediation across 1,200+ SMB 2.0, Customer Journey, and BPO tables — resolving critical compliance-risk design flaws and establishing remediation patterns later adopted at scale in the Safe Ads program
- Spearheaded migration of 1,000+ Hive pipelines to Spark, automating ~70% of conversion tasks — compressing a projected 12-month migration to under 6 months
- Owned the SMB funnel redesign, improving query performance 3x and scaling the pipeline from thousands to tens of thousands of small-business advertisers without infrastructure changes
- Enhanced Hive anonymization processes with privacy engineers, enabling ads and commerce data products to pass privacy audits without manual intervention — cutting audit preparation effort by ~60%